

Joshua (Hyun Jo) Jang

Undergraduate Student

University of Toronto, Canada

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Research & Art Portfolio: musicofmusix.github.io

Research Interests

Computational Photography, Computer Graphics, Computer-Assisted Creativity

Education

2020 ~ 2026 (Exp.) **University of Toronto**, Toronto, Canada

Honours Bachelor of Science in Computer Science and Cognitive Science

Cumulative GPA: 3.89 / 4.00; Computer Science GPA: 3.97 / 4.00

Mandatory Military Service: 2021.06 ~ 2022.12 (2 Academic Years)

Research Experience

2025.09 ~ Present **University of Toronto**, Toronto, Canada

Undergraduate Research Assistant @ Dynamic Graphics Project (DGP) Lab

Supervisors: Prof. Karan Singh, Karran Pandey

- Contributing to the development of an interactive, intelligent image brush tool for rapid scene composition and editing.
- Researching artist workflows and user experience bottlenecks.
- Conducting contextual feature extraction experiments with vision transformers, convolutional neural networks, SIFT, etc.

2025.01 ~ 2025.05 **Samsung Electronics**, Toronto, Canada

Research Intern @ AI Center - Toronto

Supervisors: Prof. Michael Brown, Dr. Hakki Karaimer

- Developed a compact, interpretable model of photographic film colour for emulation using digital images; see **Publications**. Carried out the full project pipeline with advisory input from supervisors.
- Implemented a suite of Python tools for camera ISP analysis, notably a vectorised RAW renderer, a colour chart patch value extractor, and a colour space converter.
- Constructed RAW-JPG image datasets in various environments and lighting conditions across multiple digital and film cameras for Samsung's ISP projects, involving capture, alignment and processing.

Publications

2025 **Hyun Jo Jang**, Hakki Karaimer, Michael Brown

Emulating Emulsion: A Compact Physically-Based Model for Film Colour

ACM SIGGRAPH 2025 Posters

Teaching Experience

2025.09 ~ Present **University of Toronto**, Toronto, Canada.

Undergraduate Advisor @ Computer Science Student Union

- Mentoring a small cohort of undergraduates exploring artist-driven computational research, guiding them through weekly literature reviews, research design, and project development.
- Facilitating group discussions and critique sessions to help students build confidence in expressing and pursuing research ideas.

Exhibitions

2024 **Hart House Camera Club Annual Exhibition**, Toronto, Canada.

People Photography Category Winner

2023 **Interactive Media Exhibition for Children**, Gimpo, South Korea.

Exhibitor, Artist

Creative Projects

2023 **Interactive Virtual Subway Station (“Metro”)**

- Designed and developed a stylised 3D subway station, with visuals replicating 1990s Japanese hand-drawn animation through custom-made non-photorealistic rendering (NPR) shaders in Unity.
- Installed as part of an in-person interactive exhibition for children, where attendees entering the building would be greeted with a virtual subway train arriving and departing towards the next room; see **Exhibitions**.

2022 **Aesthetics-Driven Interactive Poster/Game UI Demo (“Sairi”)**

- Developed an interactive poster/game UI demo, where the visual flair of motion graphics and the interactivity of games were combined in Unity.
- Implemented a fully animated humanoid character with expressions, 2D text and shapes in 3D space, and randomly-generated relaxed cubic splines for stylistic effects.
- Users could click or tap to switch between screens, with stylised transitions in between.

2020 ~ Present **Photography, Graphic Design, 3D Art**

- Fujifilm 2024 Workshop Toronto, Portraits
- Portfolio website design & implementation
- 3D art using custom, procedurally generated displacement maps

Skills

Programming C/C++, C#, Python, Lua

Creativity Photoshop, Illustrator, Lightroom, Blender 3D, Cinema 4D, Unity, HTML, CSS

Languages English (Fluent), Korean (Fluent)